

1 Introduction

ASSIGNMENT #01

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2025
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Due Date: Friday, September 5th TA's Mailbox by **3:00PM**

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Also place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.
- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report.

Problem A: Review of Classical Thermodynamics

Air at pressure $P_1 = 100$ kPa and temperature $T_1 = 30^\circ\text{C}$, is compressed adiabatically and isentropically to a pressure P_2 . The compressor discharge is heated isobarically in the combustor to a temperature $T_3 = 1250^\circ\text{C}$ and then expanded adiabatically and isentropically in the turbine to a pressure $P_4 = P_1$. A process flow diagram of this cycle, which is also known as the Brayton cycle, is shown in the **Figure A** below. Perform the following calculations:

1. The pressure ratio P_2/P_1 that maximizes the net power generation
2. The minimum air flow rate in MMSCFD required to produce **100 MW** of power
3. The cycle efficiency at maximum power
4. The pressure ratio P_2/P_1 that maximizes the cycle efficiency
5. The value of the maximum efficiency
6. The net power in **MW** generated at maximum efficiency
7. The Carnot efficiency of the cycle
8. If the fuel used in the combustor is kerosene what is the amount in **lbs/hr** required for maximum power production

Data and Assumptions:

1. You may assume that the air behaves as ideal gas with constant heat capacity

Figure A: Power Production: Brayton Cycle

Figure A: Power Production: Brayton Cycle

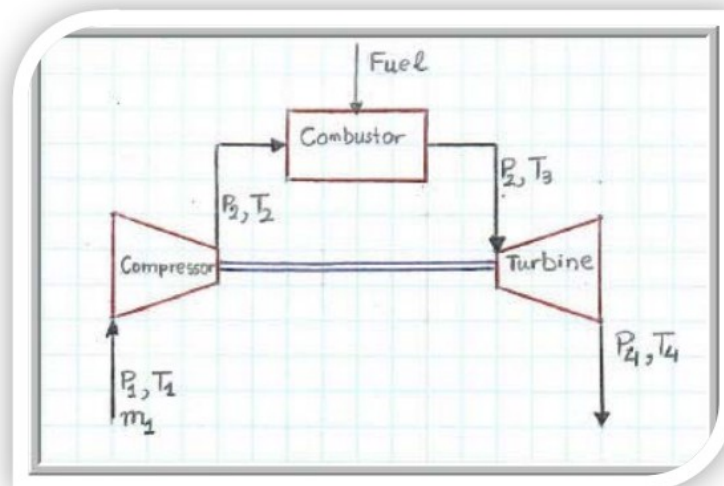


Figure 1: Brayton Cycle

A.1 Introduction to the Brayton Cycle:

Equation (1)

$$\frac{dE}{dt} = W_T - W_C$$

So the approach to this problem that we took is that we need to start with the energy balance.

Equation (2)

$$\frac{dE}{dt} = m [C_p(T_3 - T_4) - C_p(T_1 - T_2)]$$

Since the question prompts us and we are working in an adiabatic and isotropic environment, we can ignore any change in heat and entropy ($\Delta Q = 0$) and ($\Delta S = 0$) With the help of ChatGPT, we were able to figure out the relationship between Temperature and Pressure Ratio (R_p).

Equation (3)

$$R_p = \frac{P_2}{P_1} = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

From here we can do some simple algebra to get the following:

Equation (4)

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Then we multiply both sides by T_1 to isolate our temperature for T_2 to obtain our air after it is compressed, causing an increase in temperature before it is fed into our combustion chamber.

Shown by:

Equation (5)

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Then because we have an isentropic process, our temperature exiting the compressor is going to be affected. also by the pressure ratio.

This gives us:

Equation (6)

$$\frac{T_3}{T_4} = \left(\frac{P_3}{P_4} \right)^{\frac{\gamma-1}{\gamma}}$$

In addition we know that in the turbine $T_1 = T_4$. So we can change R_p as well and rewrite this as:

$$\frac{T_3}{T_1} = \left(\frac{P_3}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

And multiply T_1 over to get our expression in terms of T_3 .

Equation (7)

$$T_3 = T_1 \left(\frac{P_3}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

With these relations established we can now begin solving in terms of our W_{net} equation;

Equation (8)

$$W_{net} = \dot{m}C_p [(T_3 - T_4) - (T_2 - T_1)]$$

Then we plug in our substitutions which we solved in **Equation 5** and **Equation 7**:

$$W_{net} = \dot{m}C_p \left[(T_3 - T_3(R_p)^{-\frac{\gamma-1}{\gamma}}) - (T_1(R_p)^{\frac{\gamma-1}{\gamma}} - T_1) \right]$$

The reason that $T_3(R_p)^{-\frac{\gamma-1}{\gamma}}$ is raised to the negative power in the equation is because in **Equation 6** we have to isolate T_4 which causes us to have the inverse of the pressure ratio. Thus making the exponent negative:

Equation (9)

$$W_{net} = \dot{m}C_p [(T_3 - T_3x) - (T_1x - T_1)], \quad x = R_p^{\frac{\gamma-1}{\gamma}}$$

In the equation above we have simplified our equation for the sake of simplicity and substituted $R_p^{\frac{\gamma-1}{\gamma}}$ for x .

$$\frac{dW_{net}}{dx} = \dot{m}C_p [T_3x^{-2} - T_1]$$

We then take the derivative of both sides with respect to x and simplify we get:

$$0 = \dot{m}C_p (T_3x^{-2} + T_1)$$

To find the maximum net power of this system we have to set $\frac{dW_{net}}{dx} = 0$ Further isolating the system in terms of T_1 gets us:

Equation (10)

$$T_1 = T_3x^{-2}$$

$$x = \sqrt{\frac{T_3}{T_1}}, \quad R_p^{\frac{\gamma-1}{\gamma}} = \sqrt{\frac{1426.15 \text{ K}}{301.15 \text{ K}}} = 2.24$$

Re-substituting for x we get the following equation and when we solve for the square root we are left with R_p to the power of some gamma(γ) is equivalent to 2.24. (γ = the heat capacity ratio $\frac{C_p}{C_v}$) which through a quick Google search we discovered was 1.4

$$R_p = (2.24)^{\frac{1.4}{0.4}} = \boxed{16.86}$$

A.2 The Brayton Cycle:

$$W = \dot{m}C_p \left[\left(T_3 - T_3 R_p^{\frac{\gamma-1}{\gamma}} \right) - \left(T_1 R_p^{\frac{\gamma-1}{\gamma}} - T_1 \right) \right]$$

Recycling our substituted **Equation 8** we then set this equal to our conditions prescribed by the problem conditions that we produce **100 MW** of power.

$$100 \frac{\text{kJ}}{\text{s}} = \dot{m} \frac{\text{kg}}{\text{s}} (1.005) \frac{\text{kJ}}{\text{kg K}} [(1573.15 - 1573.15(16.82)^{-0.196}) - (303.15(16.82)^{0.296} - 303.15)]$$

From here we set

$$x = (1.005) \left(\frac{\text{kJ}}{\text{kg K}} \right) [(1573.15 - 1573.15(16.82)^{-0.196})K - (303.15(16.82)^{0.296} - 303.15)K]$$

$$100 \frac{\text{kJ}}{\text{s}} = x \left(\frac{\text{kJ}}{\text{kg}} \right) \dot{m}$$

Then by solving we get that $x = 469.603 \frac{\text{kJ}}{\text{kg}}$ and by plugging this in we get:

$$\dot{m} = 212.95 \frac{\text{kg}}{\text{s}}$$

Now in order to get this in terms of **MMSCFD**(Million Standard Cubic Feet per Day) we need to get it in terms of moles to then use as part of our Ideal gas equation.

$$\dot{n} = \frac{212.95 \frac{\text{kg}}{\text{s}}}{29.965 \frac{\text{kg}}{\text{kmol}}} = 7.158 \frac{\text{kmol}}{\text{s}}$$

Equation (11)

$$PV = \dot{n}RT$$

Then from here we need to isolate **V**(Volume) and plug in the corresponding known values:
Knowns:

$$R = 8.314 \left(\frac{\text{kJ Pa m}^3}{\text{kmol K}} \right), T = 303.15 \text{ K}, P = 100 \text{ kPa}$$

$$\dot{V} = \frac{\dot{n}RT}{P} = \frac{7.158 \frac{\text{kmol}}{\text{s}} \left(8.314 \frac{\text{kJ Pa m}^3}{\text{kmol K}} \right) (303.15 \text{ K})}{100 \text{ kPa}} = 185.30 \frac{\text{m}^3}{\text{s}}$$

With our Volumetric flow rate($\dot{V}$) in terms of $\frac{\text{m}^3}{\text{s}}$ we can then use conversions that we found on google to aide us in getting our final answer:

$$\frac{185.30 \frac{\text{m}^3}{\text{s}}}{1} \times \frac{86400 \text{ s}}{1 \text{ Day}} \times \frac{35.3147 \text{ ft}^3}{1 \text{ m}^3} \times \frac{1 \text{ MMCSFD}}{10^6 \frac{\text{ft}^3}{\text{day}}} = \boxed{565.38 \text{ MMCSFD}}$$

1.1 A.3-A.7 CHE301 HW1:

Equation (12)

3.)

$$\eta_{maxpower} = 1 - \sqrt{\frac{T_1}{T_3}}$$

With the help of Google we were able to find that in the Brayton this equation is what we need to use in order to find the cycle efficiency at maximum power.

$$\eta_{maxpower} = 1 - \sqrt{\frac{303.15 \text{ K}}{1573.15 \text{ K}}} = 0.554 = 55.4\%$$

Plugging in the corresponding known values specified from the Introduction we get our answer of $\eta_{maxpower} = 0.554$ or 55.4 percent efficient at maximum power. Which for common uses such as in jet engines isn't the most favorable for fuel and energy consumption.

Equation (13)

$$4.) R_{P_{max}} = \left(\frac{P_2}{P_1}\right)_i = \left(\frac{P_3}{P_1}\right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{T_3}{T_1}\right)^{\frac{\gamma}{\gamma-1}} = \left(\frac{1523.15 \text{ K}}{303.15 \text{ K}}\right)^{\frac{1.4}{1.4-1}} = 284.3$$

In this problem we had to consult Dr. Caracatsios to ask about how to solve. He provided us insight explaining that because the pressure will increase infinitely we had to use temperature so the temperature exiting the combustor doesn't exceed $T = 1250^\circ\text{C}$. And with a little help from Google we concluded that we had to substitute the R_p with it's corresponding temperature values as seen originally in **Equation 3** but with our values substituted in terms of T_3 and $T_4 = T_1$.

Equation (14)

$$5.) \eta_{max efficiency} = 1 - \frac{1}{R_{P_{max}}^{\frac{\gamma-1}{\gamma}}} = 1 - \frac{1}{284.3^{\frac{1.4-1}{1.4}}} = 0.8010 = 80.10\%$$

Also using Google to aid us in our process of solving we were able to find the equation to solve for the max efficiency of this cycle.

Equation (15)

$$6.) \dot{W} = \dot{m}C_p \left[(T_3 - T_3 R_p^{\frac{\gamma-1}{\gamma}}) - (T_1 R_p^{\frac{\gamma-1}{\gamma}} - T_1) \right]$$

$$\dot{W}_{max} = 212.95 \frac{\text{kg}}{\text{s}} (1.005 \frac{\text{kJ}}{\text{kg K}}) [(1523.15 - 1523.15(284.3)^{\frac{1.4-1}{1.4}})K - ((303.15)(284.3)^{\frac{1.4-1}{1.4}} - 303.15)K]$$

Plugging in our known values we are able to find the value for $\dot{W}_{max efficiency}$.

$$\dot{W}_{max} = -4.96 \times 10^{-11} \approx 0$$

This gets us a value so low that it is essentially 0 which means that when our cycle is the most efficient we generate a new power of 0 which makes sense because an ideal efficient system would generate and consume equivalent amounts of power.

$$7.) \quad \eta_c = 1 - \frac{T_1}{T_3} = 1 - \frac{303.15 \text{ K}}{1573.15 \text{ K}} = 0.801$$

This equation represents the Carnot Efficiency Equation and was also found using Google. The Carnot Efficiency Equation represents the maximum theoretical efficiency a heat engine can achieve when operation between two set temperatures.

This just confirms our previous answer for **A.5** was correct as they are both equations to find the maximum efficiency.

A.8 CHE301 HW1:

Assumptions:

1. Molar flow rate of air = 110 kg/hr
2. $\Delta W = 0$
3. Kinetic and potential energies are negligible.
4. Perfect mixing in the combustor.

After doing a DOF to figure out if the problem is solvable, doing so we end up with 11 unknown variables and 7 known variables. $DOF = 11 - 7 = 4$ and thus we had to make assumptions to further solve this problem.

$$\frac{dE}{dt} = \dot{Q} - \dot{W} + \dot{m} \left[(h_{\text{out}} - h_{\text{in}}) + \frac{V_{\text{out}}^2 - V_{\text{in}}^2}{2} + g(z_{\text{out}} - z_{\text{in}}) \right]$$

We use the energy balance equation because we need to find the molar flow rate of Kerosene going into the combustion chamber.

$$0 = \dot{Q} + \dot{m}[(h_{\text{out}} - h_{\text{in}}) + \frac{V_{\text{out}}^2 - V_{\text{in}}^2}{2} + g(z_{\text{out}} - z_{\text{in}})]$$

However after consulting with Google to save time we were able to skip the process of canceling out variables corresponding to the work, E_{Kinetic} and $E_{\text{Potential}}$ while assuming it's under steady-state conditions we are left with this equation:

Equation (16)

$$\dot{Q}_{\text{fuel}} = \dot{m}_{\text{out}}h_{\text{out}} - (\dot{m}_{\text{air}}h_{\text{air}} + \dot{m}_{\text{fuel}}h_{\text{fuel}})$$

Because the heat generated is from the combustion of the Kerosene in the air, $\dot{Q}_{\text{hr}}^{kJ}$ is in terms of the fuel we can also assume that:

Equation (17)

$$\dot{Q} = \dot{m}_{\text{fuel}} \times LHV_{\text{fuel}}$$

Thanks to CHatGPT we were able to figure out that we can write $\dot{Q}$ in terms of $\dot{m}_{\text{fuel}}$ and a value know as the Lower Heating Value(LHV).

A.8 CHE301 HW1 Continued:

This LHV assumes that water vapor does not condense in the combustion products. According to ChatGPT and Google this value is practical in uses for gas turbines and internal combustion engines, where the exhaust gases are so hot that water vapor doesn't condense. This makes it practical for our use in this problem. **Equation (18)**

$$\dot{m}_{fuel} \times LHV_{fuel} = \dot{m}_3 h_3 - (\dot{m}_2 h_2 + \dot{m}_{fuel} h_2)$$

Combining **Equation 16** and **Equation 17** we get **Equation 18**.

$$\dot{m}_{fuel} = \frac{\dot{m}_{air}(H_3 - H_2)}{H_f Kerosene + LHV_{Kerosene} - H_3}$$

Then to simplify this equation, because $LHV_{Kerosene} \gg H_3$ and $LHV_{Kerosene} \gg H_f$ we are able to ignore the effects of both in the denominator.

$$\dot{m}_{fuel} = \frac{\Delta H}{LHV_{Kerosene}}$$

Now it's time to solve for ΔH of the combustor before we use the simplified version of **Equation 18** solving for $\dot{m}_{fuel}$.

Equation (19)

$$\Delta H = \dot{m} C_p \Delta T = \dot{m} C_p (T_3 - T_2)$$

Now we substitute $T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$ into our equation and plug in the corresponding numbers to get:

$$\dot{m}_{fuel} = \frac{110 \frac{kg}{hr} 1.005 \frac{kJ}{kgK} (1523.15 - 303.15 (16.86)^{\frac{1.4-1}{1.4}}) K}{43000 \frac{kJ}{kg}}$$

Solving this gets us our answer in terms of $\frac{kg_{kerosene}}{hr}$

$$\dot{m}_{fuel} = 2.169 \frac{kg_{kerosene}}{hr} \times \frac{2.20649 lbs}{kg} = \boxed{4.786 \frac{lbs_{kerosene}}{hr}}$$

And converting this from $\frac{kg_{kerosene}}{hr}$ into $\frac{lbs_{kerosene}}{hr}$ we get our final answer in terms of lbs of kerosene per hour.

In order to double check and make sure this answer makes sense We checked on Google what the ideal mixture of Kerosene and air is in a Brayton cycle to ensure maximum power production. We found that this value at the end makes sense because the Ideal ratio of Kerosene to air is within the range of an 1:11-16 ratio.

Problem B: Numerical Solution of an Algebraic Equation.

The vapor fraction α of a two-phase mixture of hydrocarbons is given by the equation

$$F(\alpha) = \sum_{i=1}^{14} \frac{z_i (K_i - 1)}{1 + \alpha (K_i - 1)} = 0$$

where $z(i)$ is the molar composition of the mixture and $K(i)$ are thermodynamic values related to the physical properties of the hydrocarbon mixture.

First derive an expression for the derivative $\frac{dF}{d\alpha}$. Next, calculate the numerical value of the vapor fraction α using the following data for this hydrocarbon mixture:

$z(i)$	Value	$K(i)$	Value
$z(1)$	0.117598	$K(1)$	49.905777
$z(2)$	0.002775	$K(2)$	13.445753
$z(3)$	0.003435	$K(3)$	3.4749140
$z(4)$	0.003171	$K(4)$	1.2825470
$z(5)$	0.001189	$K(5)$	0.6281980
$z(6)$	0.000793	$K(6)$	0.4776770
$z(7)$	0.033971	$K(7)$	0.0644180
$z(8)$	0.044667	$K(8)$	0.0555770
$z(9)$	0.002572	$K(9)$	0.2268650
$z(10)$	0.065649	$K(10)$	0.1835740
$z(11)$	0.412276	$K(11)$	0.0927070
$z(12)$	0.032581	$K(12)$	0.1367150
$z(13)$	0.225623	$K(13)$	0.0722520
$z(14)$	0.053700	$K(14)$	0.0382340

Prepare two tables: one with the iteration sequence using the Newton method and another with the iteration sequence using the Secant method. Confirm your result by using Excel's GoalSeek method.

Hint: Familiarize yourselves and use the Excel function **SUMPRODUCT**.

Problem B Numerical Solution of the Rachford-Rice Equation

Goal

We want to find the vapor fraction $\alpha \in [0, 1]$ that satisfies the *Rachford-Rice equation*:

$$F(\alpha) = \sum_{i=1}^N \frac{z_i K_i - 1}{1 + \alpha (K_i - 1)} = 0, \quad \sum_{i=1}^N z_i = 1, \quad (1)$$

where z_i are overall mole fractions and K_i are equilibrium ratios.

Reasoning

From component balances with $y_i = K_i x_i$ and

$$z_i = (1 - \alpha) x_i + \alpha y_i,$$

we obtain

$$x_i = \frac{z_i}{1 + \alpha(K_i - 1)}, \quad y_i = \frac{K_i z_i}{1 + \alpha(K_i - 1)}.$$

The condition $\sum_i x_i = 1$ (or $\sum_i y_i = 1$) produces $F(\alpha) = 0$. Thus the single unknown α enforces normalization of phase compositions.

Monotonicity and Feasibility

Differentiating term-by-term gives

$$\frac{dF}{d\alpha} = - \sum_{i=1}^N \frac{z_i (K_i - 1)^2}{[1 + \alpha(K_i - 1)]^2}. \quad (2)$$

Since each term is negative, $F(\alpha)$ is decreasing in the feasible interval (where denominators remain positive). Therefore, if $F(0) > 0$ and $F(1) < 0$, there exists a *unique* root in $(0, 1)$.

Numerical Methods

Because F is smooth and monotone, Newton's method converges rapidly:

$$\alpha^{(k+1)} = \alpha^{(k)} - \frac{F(\alpha^{(k)})}{F'(\alpha^{(k)})}. \quad (3)$$

The secant method is also an option:

$$\alpha^{(k+1)} = \alpha^{(k)} - F(\alpha^{(k)}) \frac{\alpha^{(k)} - \alpha^{(k-1)}}{F(\alpha^{(k)}) - F(\alpha^{(k-1)})}. \quad (4)$$

Results

With the given $\{z_i, K_i\}$, all methods converge to

$$\boxed{\alpha = 0.115472700 \quad (\text{dimensionless})}. \quad (5)$$

Newton Iteration Table

Newton Method						
z	k	a	F(a)	F'(a)	Goal seek(check)	
0.117598	49.90578	0.2873	-0.67954	-2.59764	0.11547957	-5.04992E-05
0.002775	13.44575	0.0257	1.770782	-56.2447		
0.003435	3.474914	0.057183	0.706175	-20.4809		
0.003171	1.282547	0.091663	0.207024	-10.3251		
0.001189	0.628198	0.111714	0.028317	-7.71729		
0.000793	0.477677	0.115383	0.000661	-7.3622		
0.033971	0.064418	0.115473	3.73E-07	-7.35388		
0.044667	0.055577	0.115473	1.19E-13	-7.35388		
0.002572	0.226865	0.115473	6.25E-17	-7.35388		
0.065649	0.183574	0.115473	6.25E-17	-7.35388		
0.412276	0.092707	0.115473	0	-7.35388		
0.032581	0.136715	0.115473	0	-7.35388		
0.225623	0.072252	0.115473	0	-7.35388		
0.0537	0.038234	0.115473	0	-7.35388		
		Final value a:	0.11547			

Figure 2: Newton Iteration Table.

Newton's method updates the estimate of **(3)**. It uses both the function and its derivative, and for smooth monotone functions like $F(\alpha)$ it converges quadratically once close to the root. Starting from $\alpha^{(0)} = 0.5$, the method converges in about 6 steps to

$$\alpha = 0.115472700$$

Secant Iteration Table

Secant Method						
z	k	a	F(a)	F'(a)	Goal Seek(Check):	
0.117598	49.90578	0.2873	-0.67954	-2.59764	0.11547957	-5.04992E-05
0.002775	13.44575	0.2281	-0.51335	-3.09443		
0.003435	3.474914	0.045237	0.993302	-28.2361		
0.003171	1.282547	0.165794	-0.28517	-4.45041		
0.001189	0.628198	0.138904	-0.1507	-5.6486		
0.000793	0.477677	0.108766	0.051512	-8.02512		
0.033971	0.064418	0.116443	-0.0071	-7.26495		
0.044667	0.055577	0.115514	-0.0003	-7.35006		
0.002572	0.226865	0.115472	1.85E-06	-7.3539		
0.065649	0.183574	0.115473	-4.8E-10	-7.35388		
0.412276	0.092707	0.115473	-7.5E-16	-7.35388		
0.032581	0.136715	0.115473	6.25E-17	-7.35388		
0.225623	0.072252	0.115473	0	-7.35388		
0.0537	0.038234	0.115473	0	-7.35388		
		Final value a:	0.11547			

Figure 3: Secant Iteration Table.

The secant method avoids calculating the derivative directly by using a finite difference approximation between two prior iterates as shown in **(4)**. It generally converges more slowly than Newton's method, but is useful when the derivative is expensive or difficult to compute. Using initial guesses $\alpha^{(0)} = 0$ and $\alpha^{(1)} = 1$, the method converged in about 14

steps to

$$\alpha = 0.115472700$$

Excel Goal Seek

Goal Seek is a numerical root finding tool built into Microsoft Excel. It works by repeatedly adjusting the value of one input cell until a target cell achieves a specified value. In this problem, the input cell was α and the target cell was the Rachford-Rice function $F(\alpha)$. Setting $F(\alpha) = 0$ by changing α , Goal Seek automatically converged to the same solution as the Newton and Secant methods:

$$\alpha = 0.115472700$$

Phase Compositions

Finally,

$$x_i = \frac{z_i}{1 + \alpha(K_i - 1)}, \quad y_i = \frac{K_i z_i}{1 + \alpha(K_i - 1)},$$

and one may check that $\sum_i x_i = \sum_i y_i = 1$ within acceptable error.

Units

z_i , K_i , α , and $F(\alpha)$ are all dimensionless.